Varun **Nikam**

Scientist/Engineer - SC · VSSC, ISRO

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Education

Dual Degree (B.Tech. in Engineering Physics + M.S. in Astronomy & Astrophysics) *Thiruvananthapuram, India*

Indian Institute of Space Science and Technology (IIST) *Aug. 2019 – May 2024*

Cumulative Grade Point Average (CGPA): 8.11/10

Skills

Programming	Python, LaTeX, Git, GitHub, Shell scripting (bash/zsh), Make, C/C++, MATLAB, SQL, High-performance parallel computing (MPI, OpenMP)
Python Libraries	Xarray, NumPy, Astropy, SciPy, Matplotlib, Seaborn, Pandas, TensorFlow, Sphinx
Astronomical Data Analysis	HEASoft, IRAF, PyRAF, DS9
Statistical Techniques	Bayesian Analysis, MCMC, Machine/Deep Learning, Convolutional Neural Networks
Modeling/Simulation Tools	MP-GADGET, MERAXES, Venus-GRAM, Venus Climate Database, VTS3

Research Experience

Positions

Scientist/Engineer - SC *Thiruvananthapuram, India*

Vikram Sarabhai Space Centre, Indian Space Research Organisation *Oct 2024 – Present*

Projects

Synergy between future 21-cm experiments and physical cosmology: Effect of X-ray heating from Quasars on the high-redshift 21-cm signal *ANU-FRT Project, Canberra*

Advisor: Dr. Yuxiang Qin *June 2024 – Aug. 2024*

As part of the Future Research Talent (FRT) program at the Australian National University, this project aimed at exploring the impact of AGNs on high redshift 21-cm signal by forward modeling the X-ray heating in semi-analytic galaxy formation and evolution model, MERAXES. The project involved training on the use of `MERAXES`, `21cmFAST` and `DRAGONS` to run cosmological simulations and analyze simulation outputs.

Exploring growth and radiative properties of supermassive black holes at cosmic dawn using NINJA simulations *Masters Thesis, IUCAA Pune*

Advisor: Prof. R. Srianand *Aug. 2023 – May 2024*

This masters thesis project aimed to study the growth of supermassive black hole (SMBH) progenitors in cosmological simulations run using `MP-GADGET`. We investigated the evolution of emission spectrum due to accretion of gas onto SMBHs and estimate the contribution from growing seed black holes.

Supermassive Black holes at Cosmic Dawn *VSP Project, IUCAA Pune*

Advisor: Prof. R. Srianand *June 2023 – Aug. 2023*

A 7-week funded summer project to understand cosmological simulations and the numerical methods used to simulate galaxy formation and astrophysical phenomena mainly focused on black holes and implementations of sub-grid physical models. I used generated datasets to study the evolutionary growth of black holes over time via accretion and mergers.

Weather Classification Using Images *Online Internship, Spartificial*

Advisor: Vivek Singh, Sudhir Sajan *Jan. 2023 – March 2023*

In this online internship project, I present a potential weather classification method using Convolutional Neural Networks (CNN). It includes the construction of a CNN architecture constructed using `TensorFlow` and used VGG16 as a benchmark model. The performance metrics of the CNN were compared with that of the VGG16-based classifier on a publicly available weather dataset and showed that the CNN achieved comparable accuracy with the VGG16-based classifier.

Studying the spectral emission from accreting supermassive black holes

Summer Internship, IUCAA Pune

ADVISOR: PROF. R. SRIANAND — *June 2022 – Aug 2022*

As part of a credited summer internship and training project, I reviewed the standard model of accretion onto compact objects via a thin α-disk and simulated the spectrum from an accreting black hole and explored the characteristic timescales for black hole growth.

Mini-Projects

IIST, Thiruvananthapuram

AS PART OF ACADEMIC COURSEWORK — *Aug. 2022 – May 2024*

- Estimating ΛCDM cosmological model parameters from Type Ia Supernovae data using Bayesian analysis
- Simulating the evolution and dynamics of the benchmark Λ-CDM cosmological model
- Clustering in galaxies from GAMA survey data using two-point correlation function
- Estimating star formation rates in galaxies using spectral indicators
- Studying the distribution of massive stars to understand Galactic morphology
- Analyzing HII regions around stars of different spectral classes

Fellowships & Awards

- 2019–24 **IIST Merit-based Scholarship**, Sponsored by the Department of Space, Govt. of India
- 2023 **Vacation Students' Programme**, Inter-University Centre for Astronomy & Astrophysics
- 2024 **Future Research Talent Travel Award**, Australian National University

Conference Talks

Chemical Species and their Variabilities in the Venusian Atmosphere — Venus Science Conference, PRL, Ahmedabad — 2025

Workshops and Courses

Data Science in Astronomy

Pune

INTER-UNIVERSITY CENTRE FOR ASTRONOMY AND ASTROPHYSICS — *December 2023*

- 3-day workshop consisting of talks and lectures on Machine Learning, applications in astronomy and a Hackathon
- Introduced to classical ML and DL models and relevant libraries along with generative, foundation and large language models
- Learnt about real-world applications in fields of transients, gravitational waves, solar physics and more
- Participated in the final day Hackathon to collaborate with working groups to implement data science techniques in new areas

Astronomical Data Science with Python

Online

SPARTIFICIAL — *Nov. 2022 – March 2023*

- 10-week online course on applications of Python programming and machine learning in astronomy
- Learned basic Python programming, visualization techniques, image processing, data handling on astronomical datasets and machine learning using convolutional neural networks. Worked on performing feature detection and analyzing galaxy datasets.

Summer School

Online

INDIAN INSTITUTE OF ASTROPHYSICS — *July 2022*

- Introduced to various astrophysical areas of study through online lectures and hands-on sessions with insights into the latest ongoing research
- Completed programming tutorials for exoplanet detection using direct imaging, studying absorption from intergalactic medium and plotting galactic rotation curves to study dark matter

Data-Driven Astronomy

Online

UNIVERSITY OF SYDNEY (HOSTED ON COURSERA) — *August 2022*

- 6-week self-paced course focusing on computational challenges to large astronomical datasets
- Learned basic SQL querying and writing efficient Python codes to reduce time complexity for data analysis
- Implemented machine learning algorithms to classify galaxies based on their morphological features

Visualizing Filters of a Convolutional Neural Network using TensorFlow

Online

COURSERA PROJECT NETWORK — *January 2023*

Basic Image Classification with TensorFlow

Online

COURSERA PROJECT NETWORK — *January 2023*

Outreach

2024	**National Science Day Celebration Volunteer, IUCAA**, Organized various competitions for high school students at the Science and Astronomy Exhibition including a talk on *Interstellar Space Travel*	*Pune, India*
2022	**Conscientia, Technical Fest of IIST**, Organized and co-ordinated a workshop in collaboration with AstronEra on *Effective Science Communication*	*Thiruvananthapuram*
2021	**AstronEra**, Invited for a webinar on *Study and Careers in Astronomy* aimed at high school students	*Online*
2021	**Astrophysicast Podcast Series**, Interview talk on *How to be an Astrophysicist in India* covering opportunities for aspiring astronomers to pursue their interest with higher studies	*Online*